

Requirements Document (SRS)

Project Name: KuCampusBite

Date: August 1, 2025

1. INTRODUCTION

This project proposes the development of a centralized, web-based food ordering and delivery platform for the University of Karachi. The platform will provide students with a convenient and cost-effective alternative to physically traveling between departments and distant canteens. It aims to save time, reduce transportation costs, and simplify the process of ordering food, especially during peak hours.

The system will support multiple canteens with varying menus, pricing, delivery charges, and policies, while ensuring a lightweight, secure, and user-friendly experience suitable for campus use. It will also assist canteen staff in managing orders efficiently, helping to reduce crowding and long waiting times. Additionally, the platform will offer a free delivery service to make food access more affordable and accessible for all users.

1.1 PURPOSE

The purpose of this project is to improve the accessibility, speed, and convenience of food services within the University of Karachi. At present, students face challenges such as long queues, crowded canteens, and limited access to food from other departments due to distance. These issues result in wasted time and reduced overall satisfaction with campus dining options.

The proposed system will address these challenges by enabling users to place orders online from any canteen on campus and have them delivered free of charge to their location. For canteen staff, it will provide a structured order management system to handle incoming requests efficiently, thereby reducing operational strain and overcrowding. Ultimately, the platform aims to create a faster, more organized, and more accessible food service environment for the entire university community.

1.2 SCOPE

The proposed system is a centralized, web-based food ordering and delivery platform for the University of Karachi, designed to streamline the process of ordering meals from multiple on-campus canteens. The platform will prioritize accessibility, ease of use, and operational efficiency for both students/teachers and canteen staff.

Included Features:

1. **Meal Pre-Ordering, Scheduling, and Reservation** – Users will be able to place orders in advance, schedule a delivery time, and reserve meals to ensure availability during peak hours.
2. **Secure User Login and Basic Profile Management of Canteens** – Each canteen will have a secure login to manage its own profile, including menu details, operating hours, and delivery policies.
3. **Canteen Dashboard for Managing Orders and Menus** – A dedicated interface for canteen staff to view, update, and process incoming orders efficiently, as well as modify menu items when needed.
4. **Mobile-Responsive Design for Student Accessibility** – The platform will be optimized for mobile devices to allow students to place orders easily from anywhere on campus.
5. **Initial Training and Onboarding of Canteen Staff** – Basic training will be provided to canteen staff to ensure they can effectively use the system for order management and customer service.

Excluded Features:

1. **Payment Gateway Integration** – The initial version will not include online payment options; all transactions will be processed via cash on delivery.
2. **Integration with Student ID Cards or University Systems** – The platform will operate independently without linking to existing university databases or student ID systems.
3. **Off-Campus Deliveries** – The delivery service will be limited strictly to locations within the university campus.
4. **Multilingual Interface** – The platform will initially be available in English only, with potential for additional languages in future updates.
5. **Mobile Application Development** – The system will be delivered strictly as a responsive website, with no downloadable mobile app in the initial release.
6. **Real-time Tracking** – The initial release will not include a live update feature showing the current status of a food order; this may be introduced in future versions. Only an estimated delivery time will be shown.

1.3 DEFINITION, ACRONYMS, ABBREVIATIONS

Platform – The centralized web-based food ordering and delivery system.

User – Any student using the platform to place food orders.

Canteen Staff – Employees responsible for preparing and delivering food orders.

Delivery Service – The process of transporting prepared meals from canteens to users’ locations within the university.

UoK – University of Karachi.

UI – User Interface.

1.4 REFERENCES

- Business Case Document – Provides the initial problem statement, project objectives, and feasibility analysis forming the foundation for the KU Campus Bite system.
- Project Charter – Defines the project scope, identifies stakeholders, outlines key deliverable, and specifies project constraints.
- Project Plan – Details the project timeline, milestones, resource allocation, and execution strategy.
- Online Research – Includes industry best practices, relevant case studies, and reference materials supporting the system requirements specification.

1.5 OVERVIEW

The proposed system is a centralized, web-based platform designed to simplify and improve the process of ordering and delivering food within the University of Karachi. The platform will cater to students, offering a convenient alternative to traveling between departments and distant canteens. By integrating multiple canteens into a single interface, the system will allow users to view menus, place orders, and schedule deliveries without physically visiting the canteens.

The platform will be mobile-responsive to ensure accessibility on smartphones, tablets, and desktop computers. Canteen staff will be provided with a dedicated dashboard for managing menus, processing orders, and tracking deliveries. While the initial version will not include real-time order tracking, users will be shown an estimated delivery time to set expectations. The system will operate on a cash-on-delivery model and will only cover deliveries within the university campus.

Ultimately, the platform aims to save time, travel cost, reduce congestion in canteens, improve operational efficiency for staff, and make food access quicker and more convenient for the university community.

2. Overall Description

2.1 PRODUCT PERSPECTIVE

This project is a new, centralized web platform that connects all university canteens. It replaces the current manual system of physically visiting canteens and waiting in queues. The system will

work as a central hub where students can browse menus, order food, and get it delivered anywhere on campus.

2.2 PRODUCT FUNCTIONS

- Show menus from all canteens.
- Allow users to place and schedule orders.
- Track orders in real time.
- Rate food and delivery service.
- Enable canteens to manage menus and stock.
- Allow canteen staff to accept and update orders.
- Provide delivery management for campus riders.

2.3 USER CLASSES AND CHARACTERISTICS

- Students & Staff (Users): Order and track food; rate service.
- Canteen Staff: Manage menus, process orders, update availability.
- Delivery Team: Pick up and deliver food.
- University Admin: Monitor performance and manage operations.

2.4 OPERATING ENVIRONMENT

- Web-based application accessible on computers, tablets, and mobile phones.
- Requires internet access (Wi-Fi or mobile data).
- Works on modern browsers (Chrome, Firefox, Edge, Safari).

2.5 DESIGN AND IMPLEMENTATION CONSTRAINTS

- Must be accessible only within the university network for security.
- Real-time updates needed for menus and order status.
- Budget and development time constraints.
- Needs simple interface for all user types.

2.6 USER DOCUMENTATION

- User Guide: Step-by-step instructions for placing orders, tracking, and rating service.
- Admin Guide: Instructions for managing menus, orders, and delivery schedules.
- Training Materials: Simple video tutorials or physical training for new users and FAQs.

3. Specific Requirements

3.1 FUNCTIONAL REQUIREMENTS

1. Authentication & Access

- User Login – Users shall be able to log in using a valid email and password.
- Admin Login – Administrators shall be able to log in using a valid email and password.
- Guest Access – The system shall allow users to skip login and place orders without creating an account.

2. User Functions

- Menu Browsing – The system shall display menus for all registered canteens, including item name, description, price, and availability status.
- Search – Users shall be able to search for menu items by name.
- Cart Management – Users shall be able to add items to a cart, adjust quantities, and remove items before checkout.
- Order Placement – Users shall be able to confirm and submit orders from the cart.

3. Canteen Staff Interaction

- Order Notification – The system shall send order details to the respective canteen staff via WhatsApp.
- Menu Availability Updates – The system shall allow canteen staff to update item availability via a designated Google Form linked to a Google Sheet.

4. Admin Functions

- Order Management – Admins shall be able to view all orders placed in the system.
- Canteen Management – Admins shall be able to add, edit, or remove canteen profiles.
- Menu Management – Admins shall be able to add, edit, or remove menu items for each canteen and update availability.

5. Data Management

- Data Storage – The system shall store user, canteen, menu, and order data in a database for retrieval and reporting.

Planned Features

- a. Order Scheduling – Users may be able to schedule orders for a future time slot.
- b. WhatsApp Order Status Updates – Users may receive order status notifications via WhatsApp based on canteen staff updates.
- c. Feedback & Ratings – Users may submit feedback or rate orders/canteens.

3.2 NON-FUNCTIONAL REQUIREMENTS

1. Performance

- The system should load the menu page within 3–5 seconds on a stable internet connection.
- Each food item search query should display results within 3 seconds.

2. Availability

The system should be available for use at least 90% of the time during testing and demonstration phases.

3. Usability

First-time users should be able to place an order without external guidance within 5 minutes of using the system.

4. Security

- User passwords should be stored in hashed form in the database.
- Basic input validation should prevent at least 90% of common invalid inputs (e.g., empty fields, wrong formats).

5. Maintainability

The codebase should follow a modular structure where at least 70% of the core functions can be updated without affecting other modules.

6. Compatibility

The system should be accessible on at least 2 major browsers (Chrome and Firefox) and 2 device types (mobile and desktop).

7. Data Accuracy

Menu items and prices displayed should match the database records with 100% accuracy during testing.

3.3 INTERFACE REQUIREMENTS

1. User Interface Requirements:

- The system shall provide a responsive UI that adapts seamlessly to mobile, tablet, and desktop screens.
- Menu items shall be displayed as image cards showing the name, price, and an “Add to Cart” button.
- The interface shall support search functionality and category filters for easy menu browsing.
- Users shall be able to view their cart contents and total price prior to confirming an order.

2. Hardware Interface Requirements:

- The system shall support access from standard smartphones and computers with internet connectivity.
- No specialized hardware shall be required for users or canteen staff beyond a web-enabled device.
- Canteen staff shall receive orders via the WhatsApp application on their smartphones.

3. Software Interface Requirements:

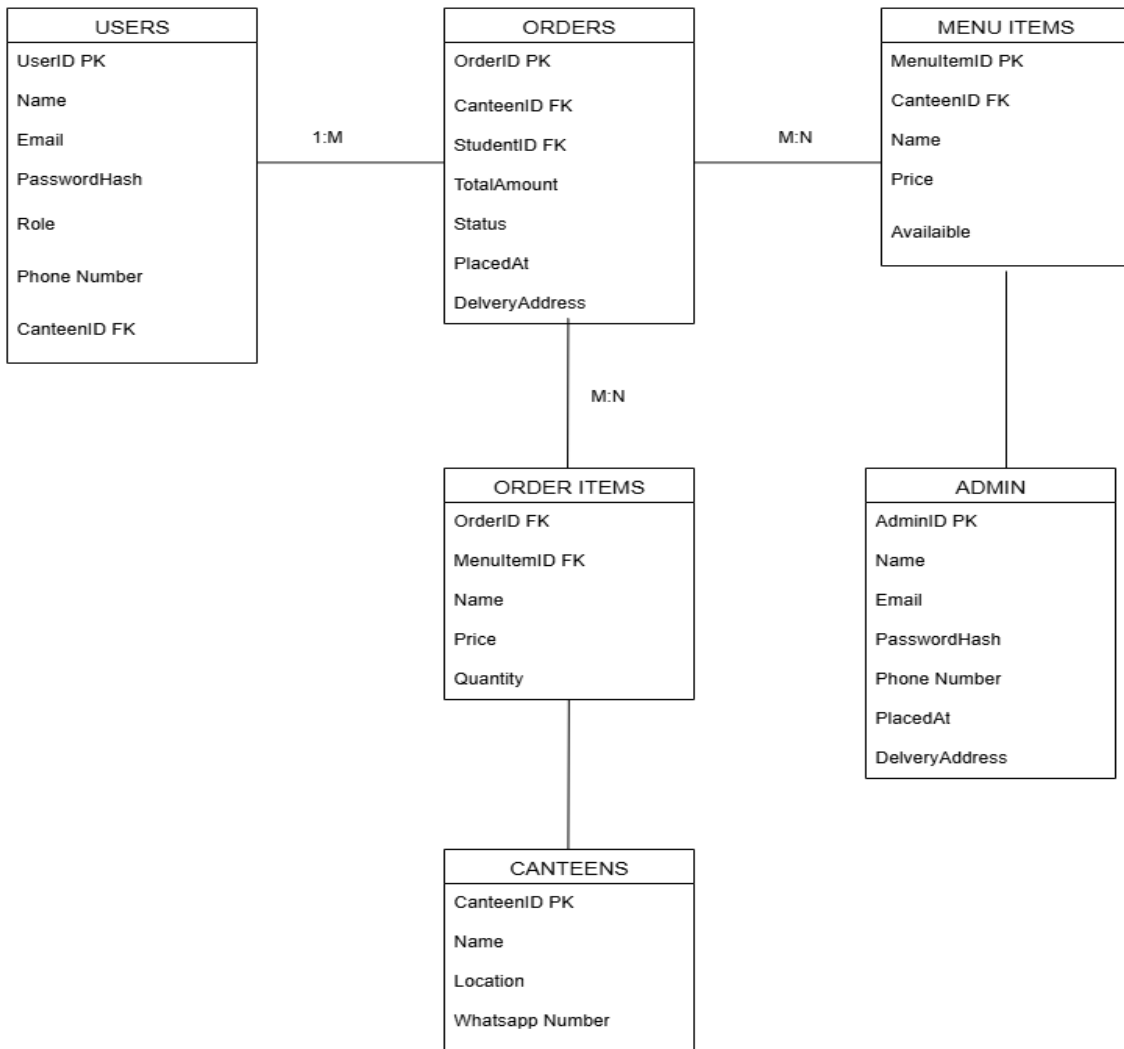
- The frontend shall be developed using HTML, CSS, and JavaScript.
- The backend shall be implemented using Node.js with the Express.js framework.
- MongoDB Atlas shall be used as the database service.
- The system shall integrate with the WhatsApp API for order notifications.

4. Communications Interface Requirements:

- Data exchange between frontend and backend shall occur over HTTPS using RESTful APIs.
- Order notifications shall be sent to canteen staff via the WhatsApp API, leveraging encrypted messaging.
- JSON shall be used as the data format for all API requests and responses.
- The system requires a stable internet connection for reliable operation.

4. System Models

4.1 Entity Relationship Diagram



4.2 Use Case Diagram

